PURNIMA BANK

SentinelAI Fraud Detection Platform

FUNCTIONAL REQUIREMENTS DOCUMENT

SentinelAI Fraud Detection Platform

Functional Design – Real-Time Scoring, Case Management & Monitoring

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Related Document:	BRD v1.2

1. Purpose

This Functional Requirements Document (FRD) translates the approved Business Requirements Document (BRD) into detailed functional specifications for engineering, data science, and QA. Each functional requirement (FR) traces back to one or more business requirements (BR).

2. System Architecture Overview

SentinelAI is composed of four logical components: (1) a real-time scoring service invoked synchronously by the card authorization switch, (2) a Snowflake-based feature store and offline training pipeline, (3) a case management integration layer, and (4) a monitoring and model governance layer. Transactions are scored in-line; the authorization decision itself remains owned by the existing switch, which consumes the SentinelAI score as one input alongside existing rules.

3. Functional Requirements

3.1 Real-Time Scoring Service (traces to BR-01, BR-02, BR-03)

ID	Functional Requirement	Traces to
FR-01	Expose a synchronous REST API endpoint (POST /v1/score) accepting a transaction payload and returning a fraud score (0–999), decision recommendation, and reason codes.	BR-01
FR-02	API shall respond within 200ms at P95 and 350ms at P99, measured at the service boundary.	BR-02
FR-03	Score thresholds shall be configurable via an admin console, segmented by card product (credit/debit) and MCC group, without requiring redeployment.	BR-03
FR-04	The service shall fail open to the legacy rules engine if scoring latency exceeds 400ms or the service is unavailable, logging a fallback event.	BR-02

3.2 Feature Store & Model Pipeline (traces to BR-01, BR-04)

ID	Functional Requirement	Traces to
FR-05	Feature pipeline shall compute rolling transaction velocity features (1hr/24hr/7-day counts and amounts) per card and per merchant, refreshed in Snowflake streams.	BR-01
FR-06	Feature pipeline shall incorporate device fingerprint, IP geolocation, and distance-from-home-address features where available from the digital channel.	BR-04
FR-07	Feature store shall support point-in-time correct joins to prevent training/serving skew and label leakage.	BR-01
FR-08	Model training pipeline shall support champion/challenger evaluation with configurable traffic splits.	BR-01

3.3 Case Management Integration (traces to BR-06, BR-07, BR-08, BR-09)

ID	Functional Requirement	Traces to
FR-09	Scores above the configured threshold shall auto-create a case in the case management system within 5 seconds via API, including the top-5 SHAP-based reason codes.	BR-06, BR-07
FR-10	Reason codes shall be mapped to a business-readable taxonomy (e.g., “unusual location”, “velocity spike”) rather than raw feature names.	BR-07
FR-11	Analyst dispositions (confirmed fraud / false positive / unable to determine) shall be written back to a labeled events table for retraining.	BR-08

3.4 Governance, Audit & Monitoring (traces to BR-10 – BR-15)

ID	Functional Requirement	Traces to
FR-12	Every scoring event shall be persisted with model version, input features, score, and decision for a minimum 7-year retention period.	BR-13
FR-13	A model card shall be maintained per model version documenting training data, features, performance metrics, and known limitations.	BR-11
FR-14	An automated fairness check shall run pre-deployment comparing false-positive rates across demographic proxies (zip-code income tier) with alerting on disparity >10%.	BR-12
FR-15	A monitoring dashboard shall track daily precision, recall, PSI (population stability index), and case volume, with automated alerts on threshold breach.	BR-14, BR-15

4. Non-Functional Requirements

Category	Requirement
Availability	99.95% uptime for the scoring service (aligned to card authorization SLA)
Scalability	Support peak load of 1,200 transactions/second (Black Friday / holiday peak)
Security	All data encrypted in transit (TLS 1.2+) and at rest (AES-256); PCI-DSS scope applies
Auditability	Full lineage from raw transaction to model decision, queryable within 24 hours for regulatory requests

5. Data Requirements

Source systems include the core banking transaction feed, the digital banking session log (device/geolocation), and the existing case management system. All feature engineering occurs in Snowflake; a dedicated FRAUD_ANALYTICS database with role-based access control isolates PII per the Bank's data classification policy.

6. Traceability Summary

All 15 functional requirements trace to the 15 business requirements defined in the BRD. No BRD item is without at least one corresponding FR. This traceability matrix will be extended into the Test Plan for UAT case coverage.

7. Approval

Name / Role	Approval	Date
Priya Nataraj – Head of Data Science	Approved	Feb 20, 2026
Marcus Webb – Head of Card Payments	Approved	Feb 21, 2026
Mautik Patel – Project Delivery Manager	Approved	Feb 21, 2026